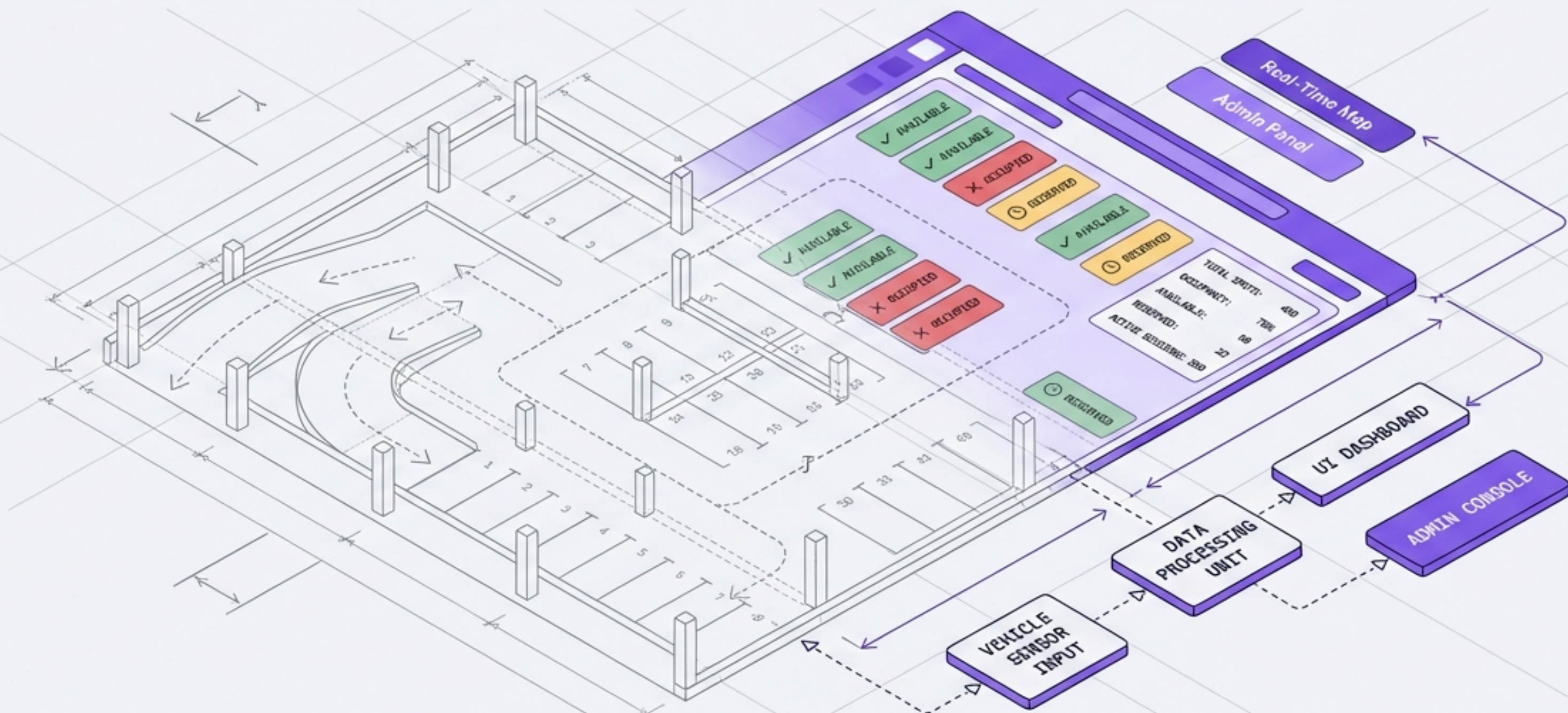


ParkFlow: Smart Parking Management

Real-time spot tracking, presence-aware availability, and seamless administration.



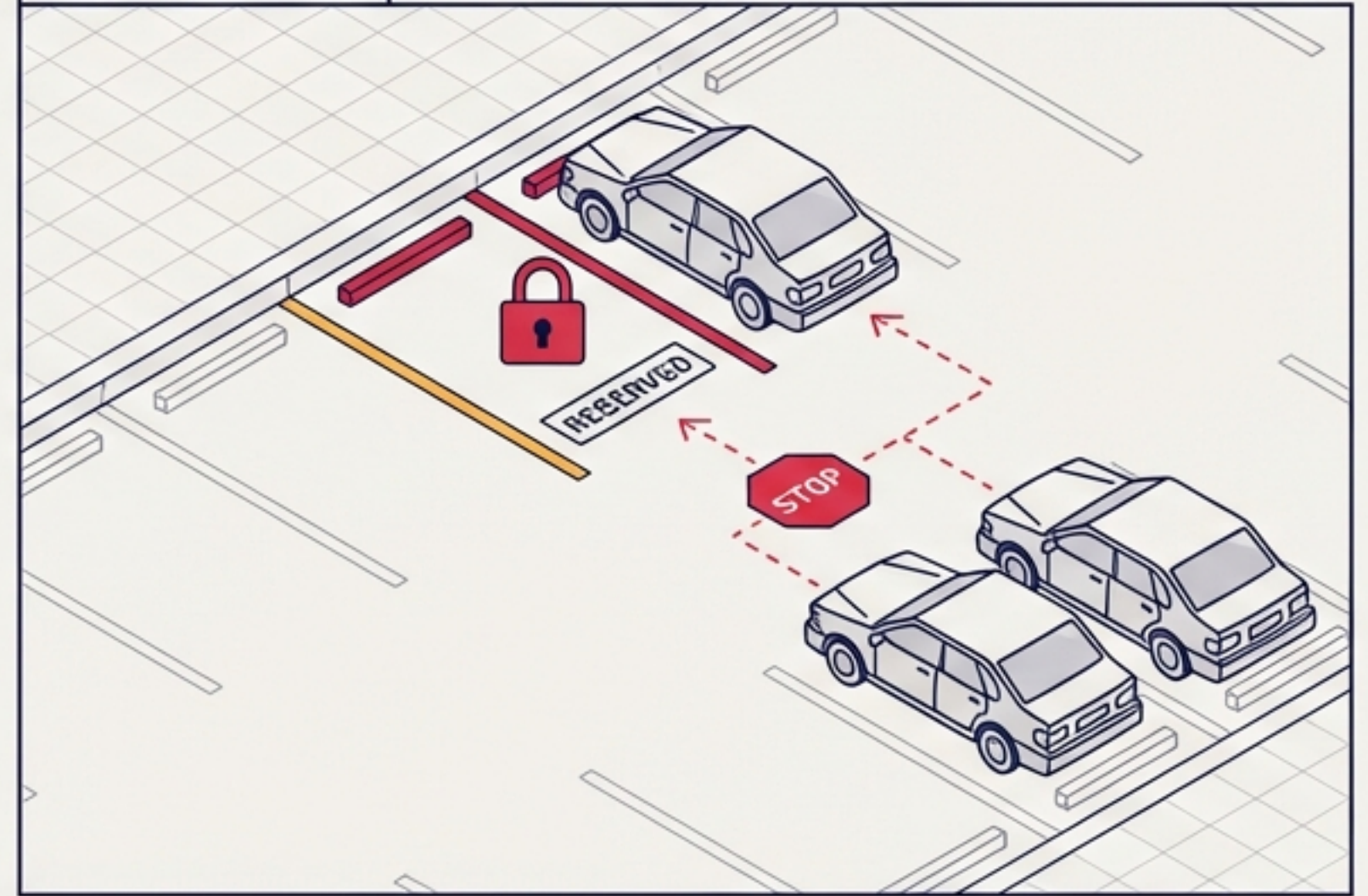
A flawless operational tool built on an uncompromising technical foundation.

Artificial Scarcity in the Physical World

The Roster

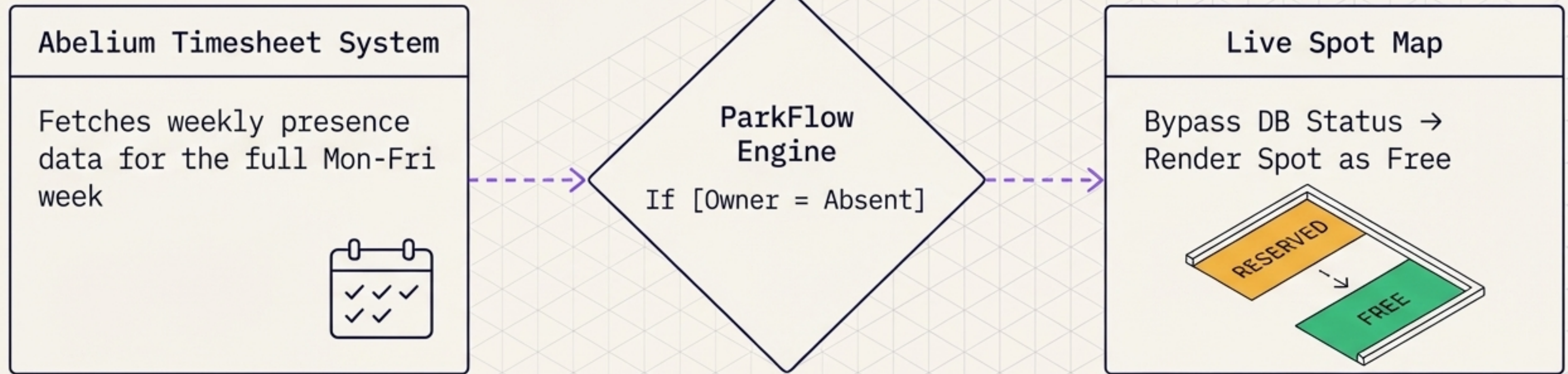
Employee's calendar							Today ▾	>
Sun	Mon	Tue	Wed	Thu	Fri	Sat		
20	31	1	2	3	4	5		
					Available	Available		
Out of Office / Remote								
					10	29		
						Available		
20	21	23	24	25	27	29		
	Available		Booked meeting 2:26 - Ops Reserved	Booked meeting 4:20 - Sps Reserved	Booked meeting 3:36 - Sps Reserved	Booked meeting 3:26 - Sps Reserved		
27	28	30	31					
available	Available	Booked meeting 3:30 - Sps Reserved	Booked meeting 3:30 - Sps Reserved					

The Reality



Reserved spots sit empty when owners are working remotely or on leave, creating friction and artificial scarcity for colleagues.

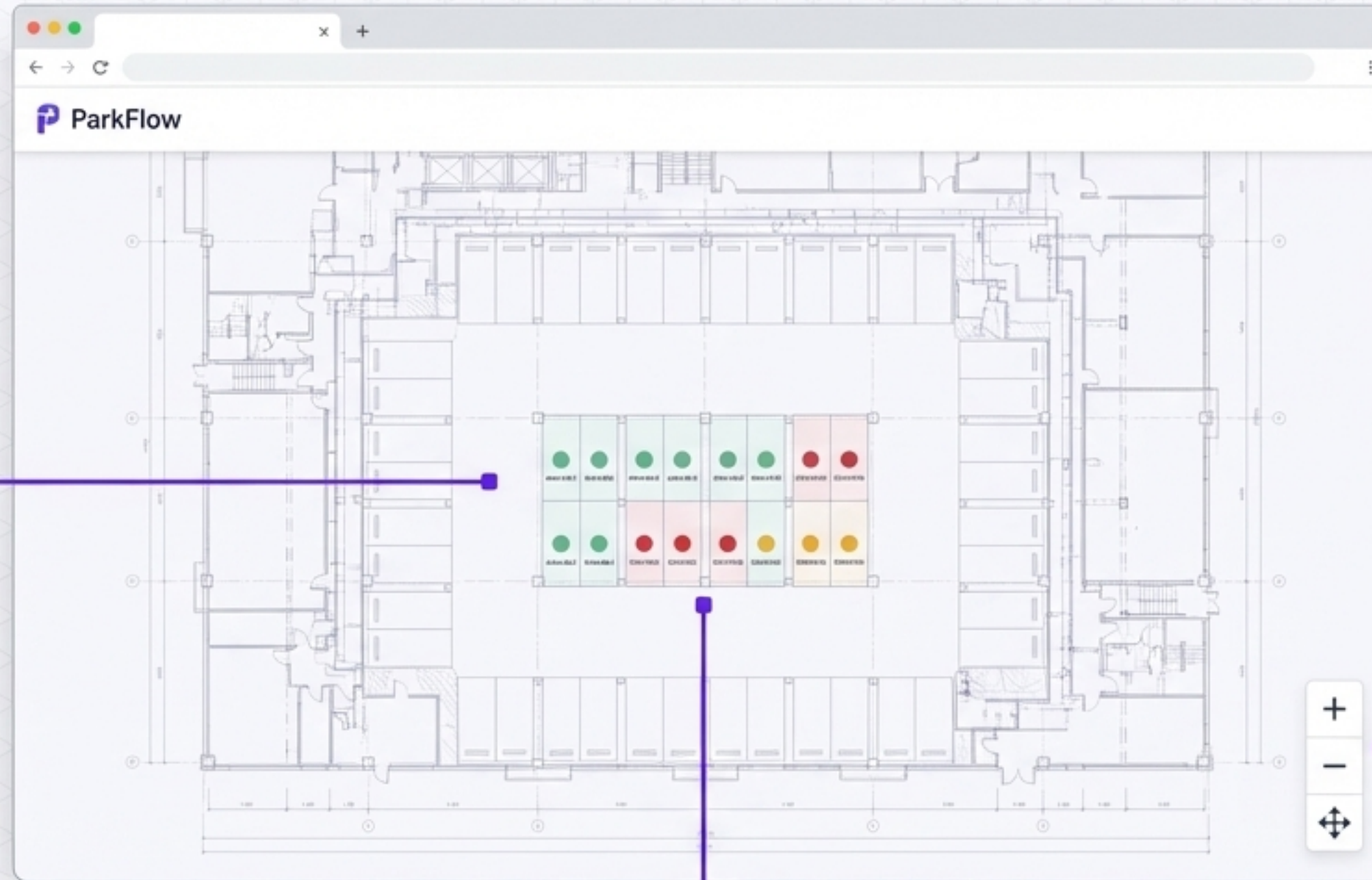
The Core Innovation: Presence-Aware Availability



Complete automation. The system merges timesheet data client-side. When an owner is absent, their spot automatically unlocks for the day—returning to reserved tomorrow.

The Interface: Interactive Digital Maps

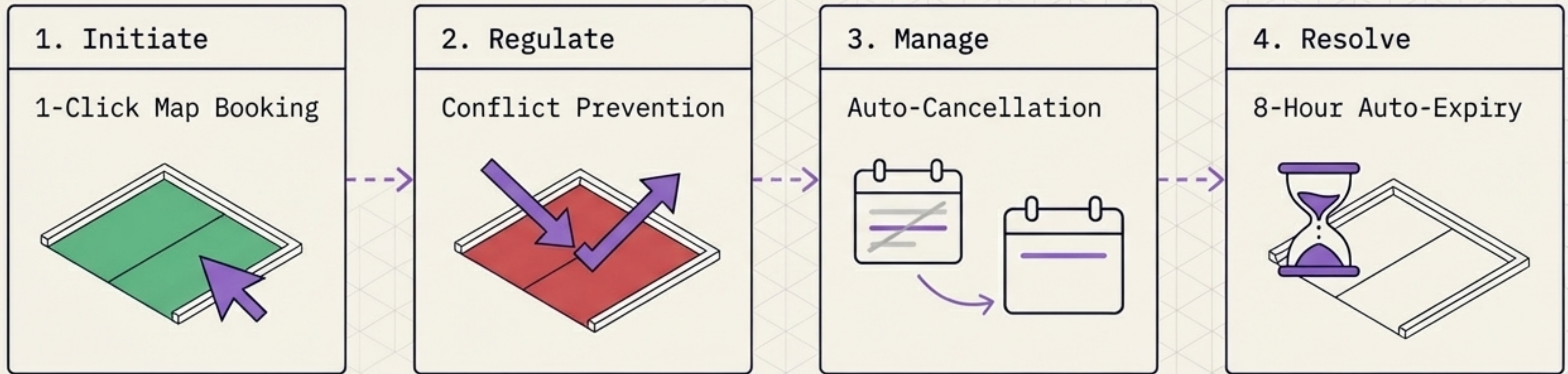
Real CAD Floor
Plan Overlays



Pinch-to-Zoom &
Pan Navigation

Live Status Indicators

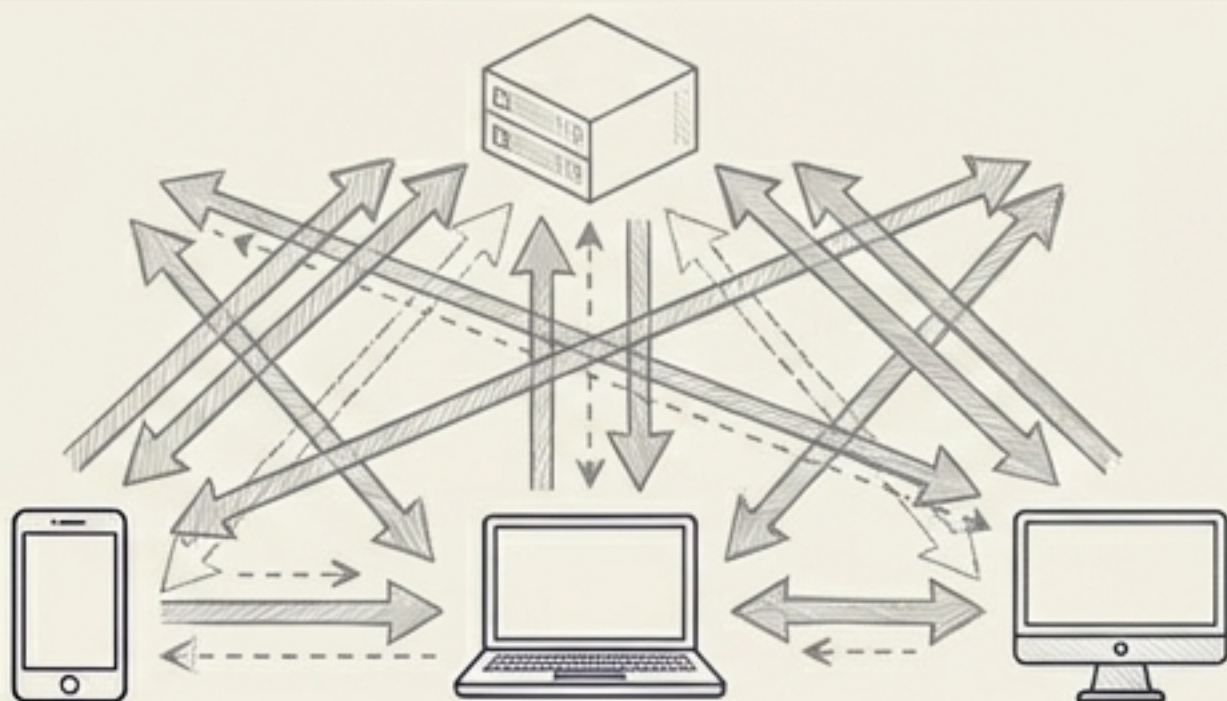
The Booking Engine: Self-Regulating and Secure



Built-in owner protection ensures spot owners who book their own spaces cannot be overridden by administrators.

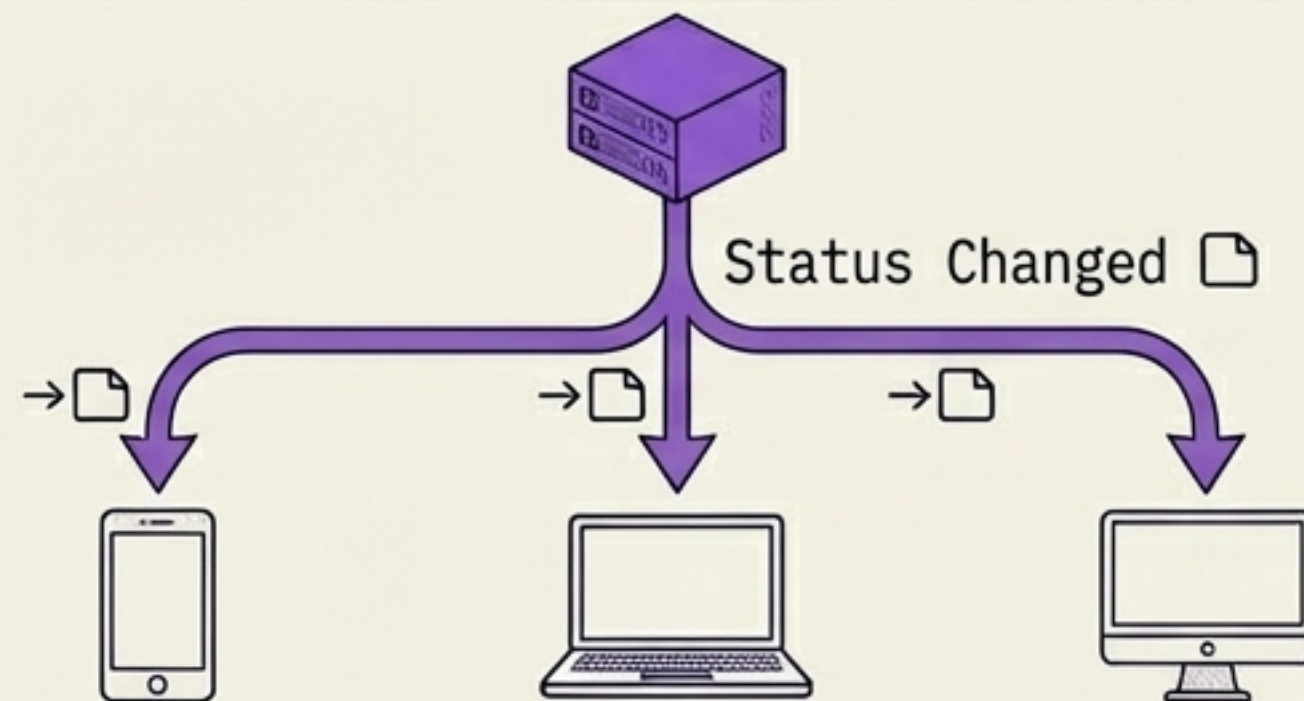
The Engine of 'Now': Server-Sent Events (SSE)

Traditional Polling - The Old Way



Heavy Load, Noticeable Lag

Server-Sent Events - **The ParkFlow Way**



Instant Broadcast, Zero Polling

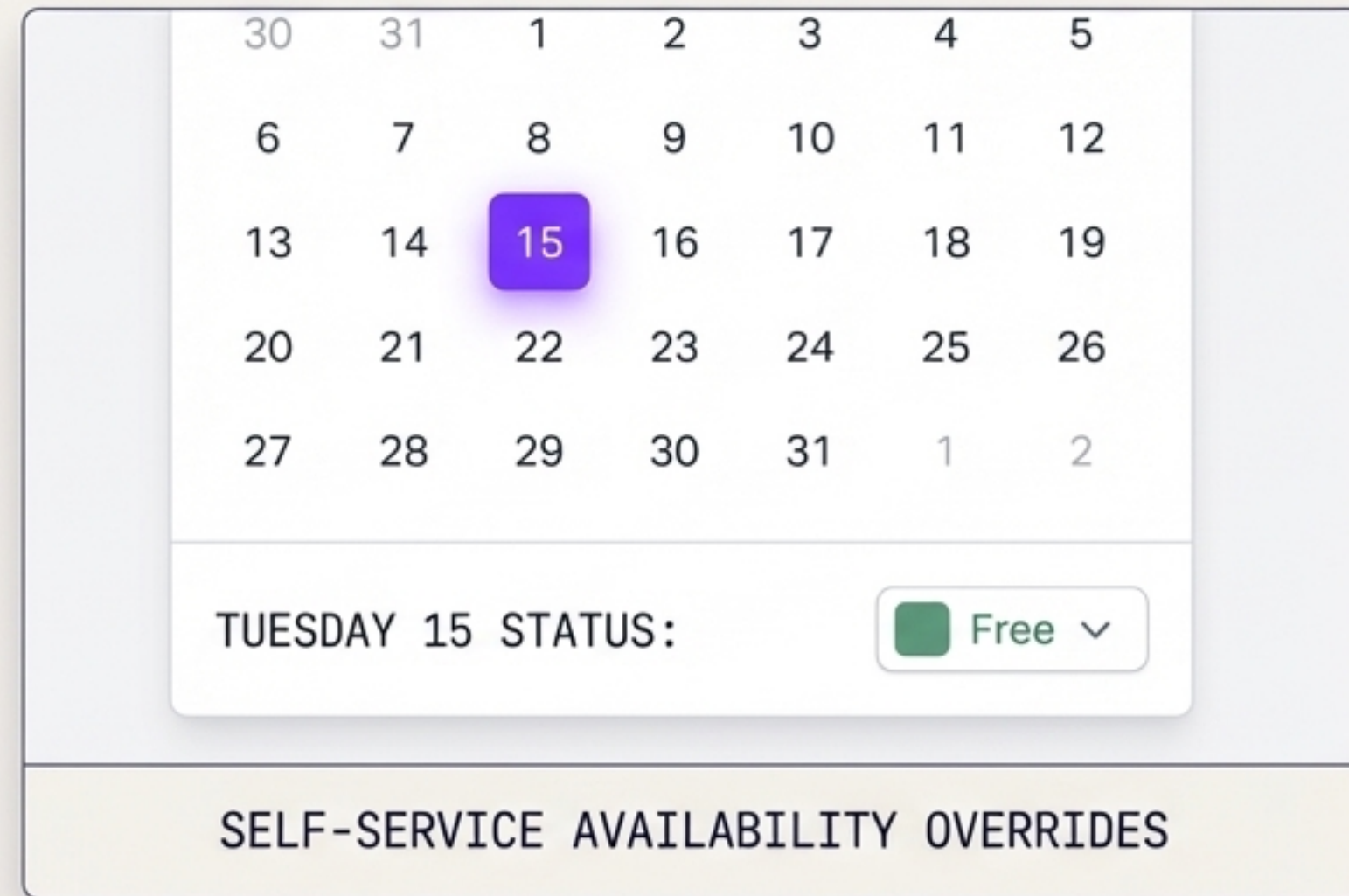
Live updates without the lag. Clients subscribe to `/api/subscribe` for instant broadcast changes. Zero polling required.

The User Ecosystem Matrix

	Everyday User	Spot Owner	Administrator
Book free spots	✓	✓	✓
Submit feedback/bugs	✓	✓	✓
View weekly booking timeline	✗	✓	✓
Set per-day spot overrides	✗	✓	✓
Visual Map Editor & CRUD	✗	✗	✓

Spotlight: Granular Control Portals

Owner Portal



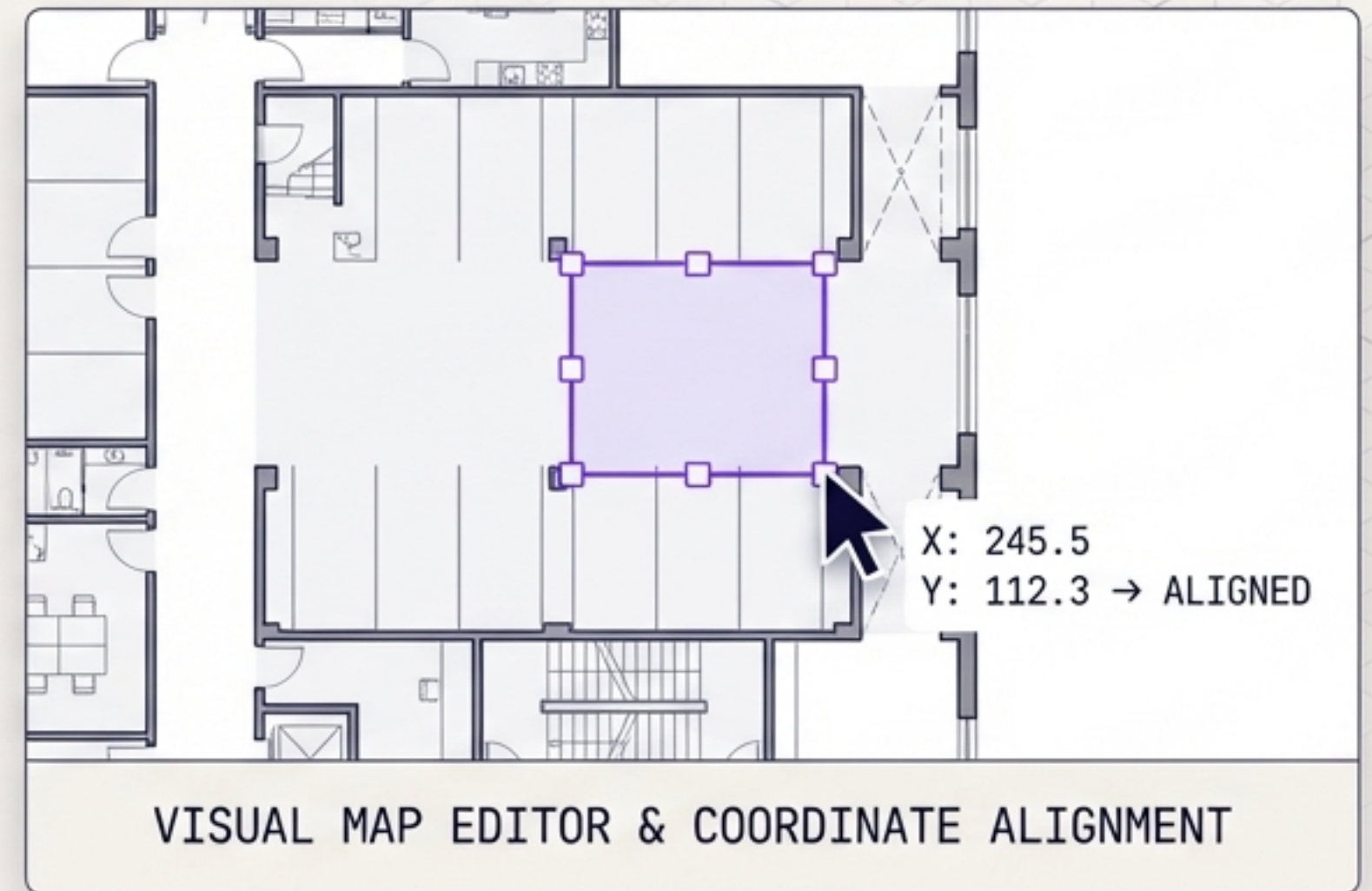
The Owner Portal interface features a calendar for managing availability overrides. The calendar shows dates from 30 to 2. The date 15 is highlighted in purple. Below the calendar, the status for Tuesday 15 is shown as 'Free' with a green square icon and a dropdown arrow. The bottom of the portal is labeled 'SELF-SERVICE AVAILABILITY OVERRIDES'.

30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2

TUESDAY 15 STATUS: Free ▾

SELF-SERVICE AVAILABILITY OVERRIDES

Admin Portal



Spot owners gain self-service availability overrides, whilst administrators get powerful visual tools to reposition coordinates directly onto the floor plan image, persisting instantly to the database.

Enterprise-Grade Security & Accountability

Access Control

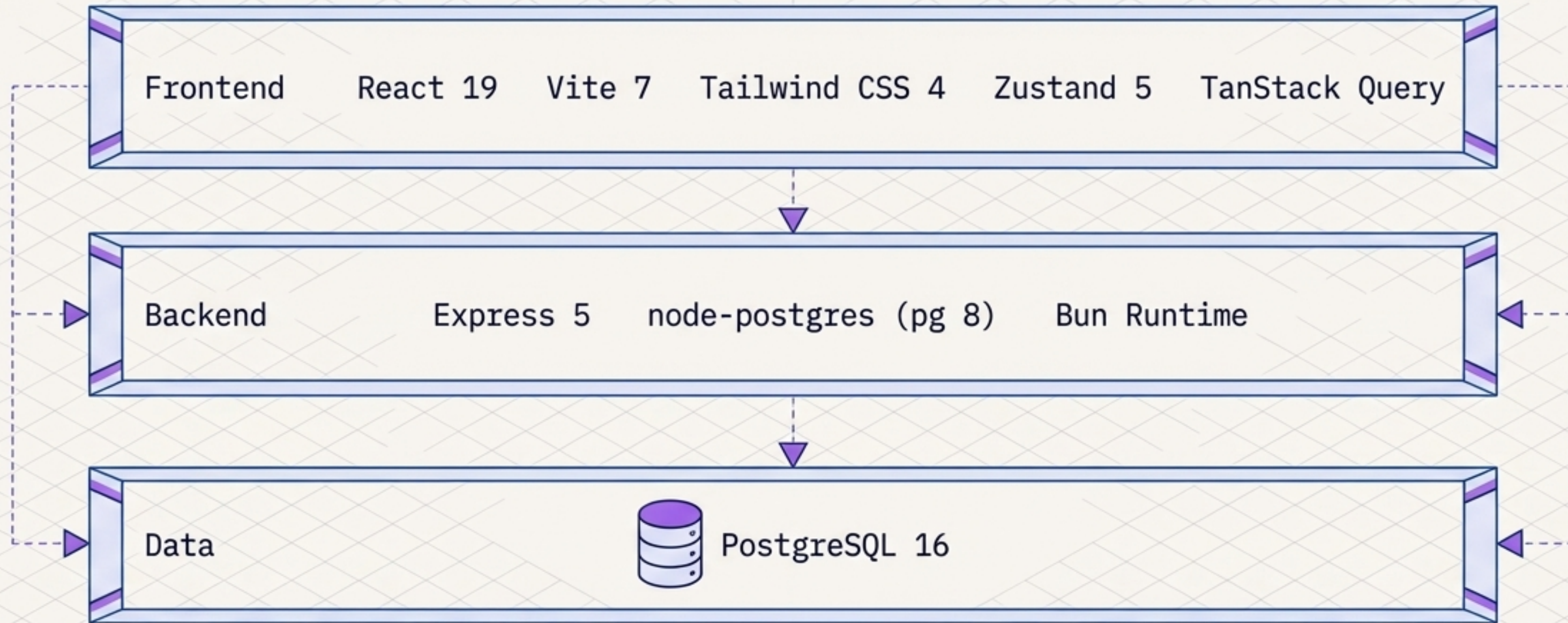


Authentik SSO via OAuth 2.0 / PKCE
Seamless password-less entry
Strict server-side token validation
Automated admin role mapping

Accountability Ledger

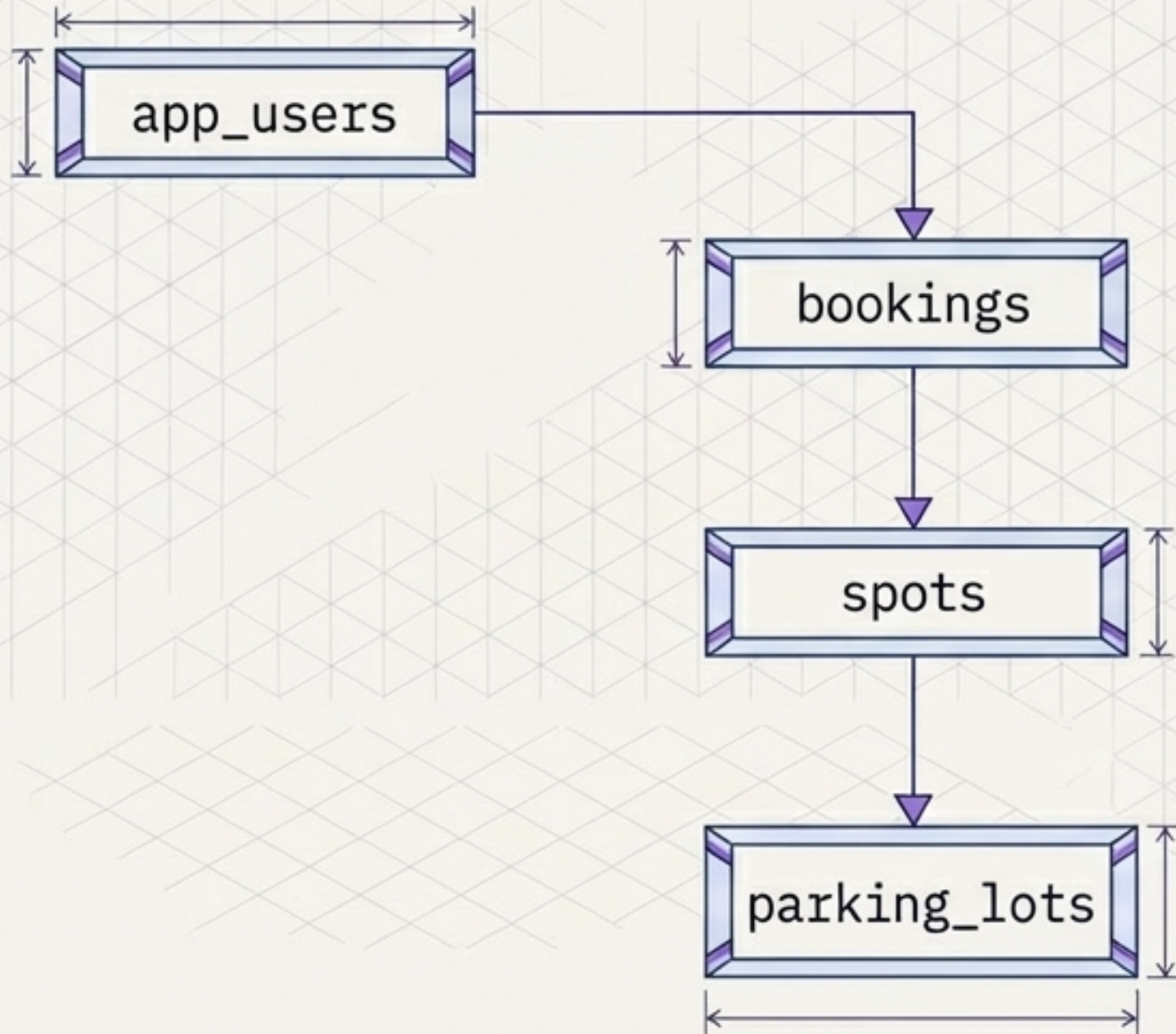
spot_changes audit log
Immutable tracking of status changes
Records every ownership reassignment
Captures **absolute before** and **after** values

Under the Hood: The Modern Tech Stack

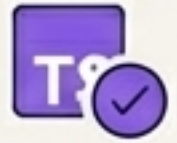


Built for absolute speed, strict type safety, and real-time performance
using the latest generation of web tooling.

Data Architecture & Quality Craft



Strict TypeScript
(No 'any' types allowed)



Mobile-First UX
(100% Tailwind utility classes)

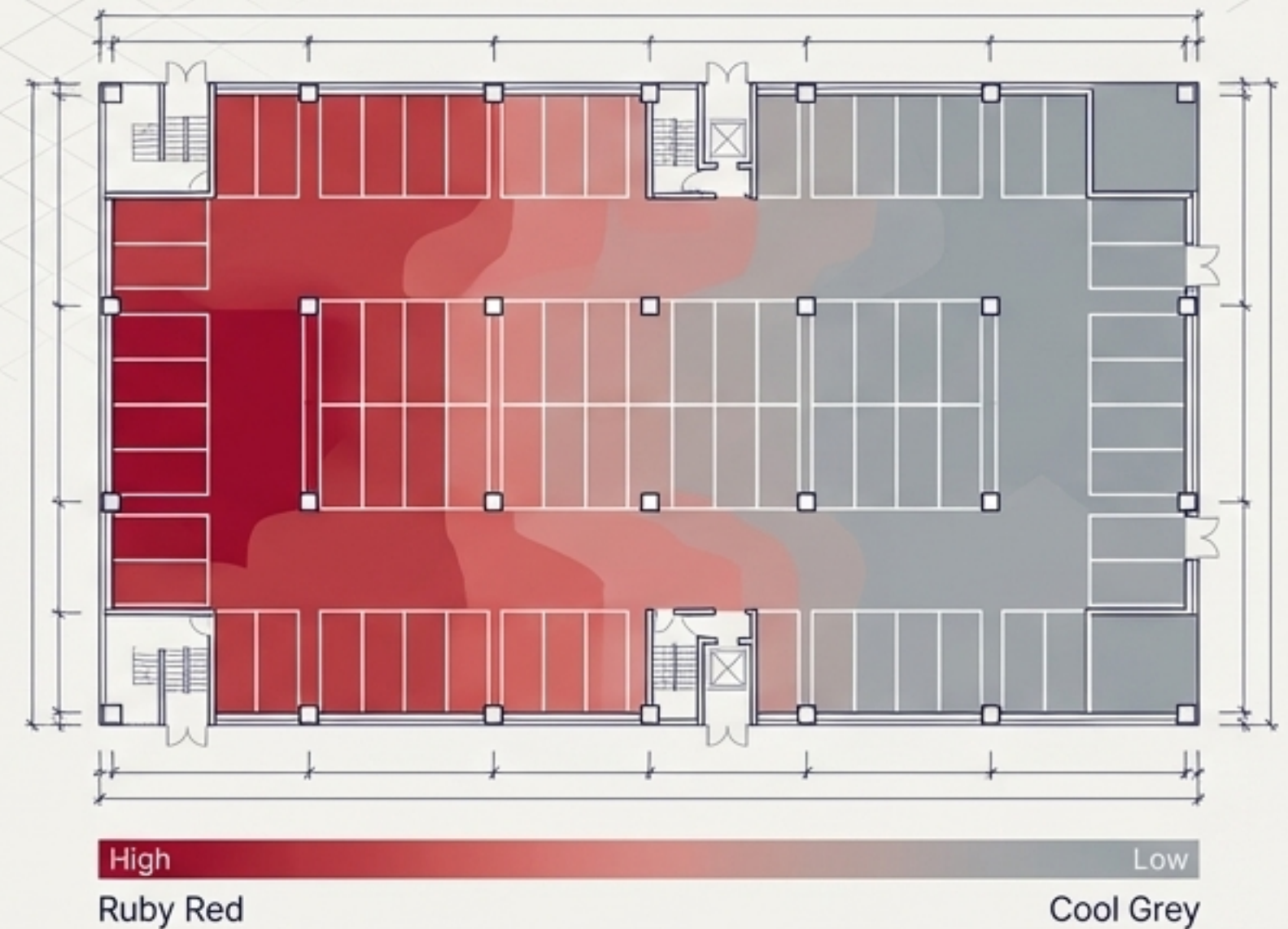
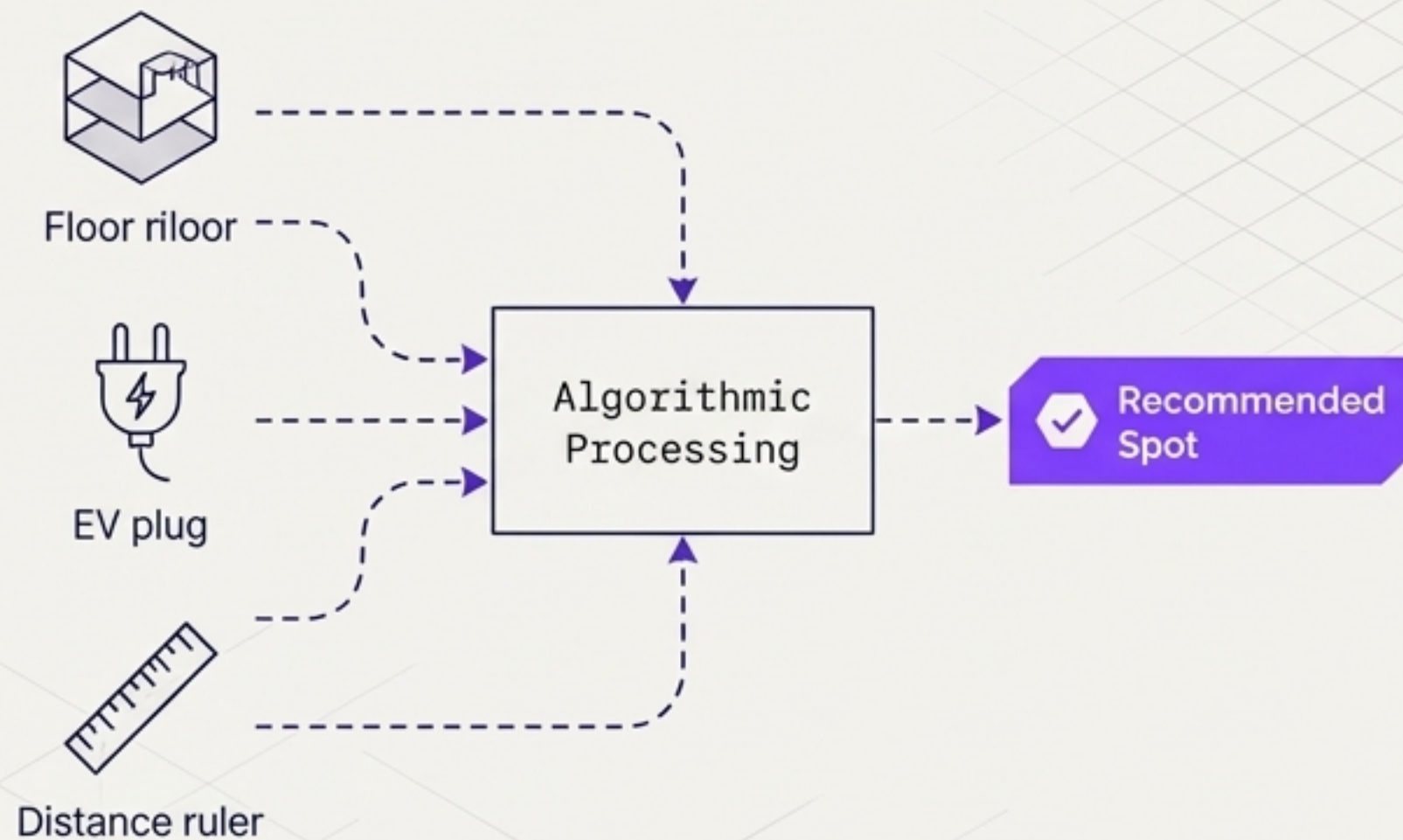


 **Algorithmic Consistency**
(Claude Code system.md
UI source of truth)

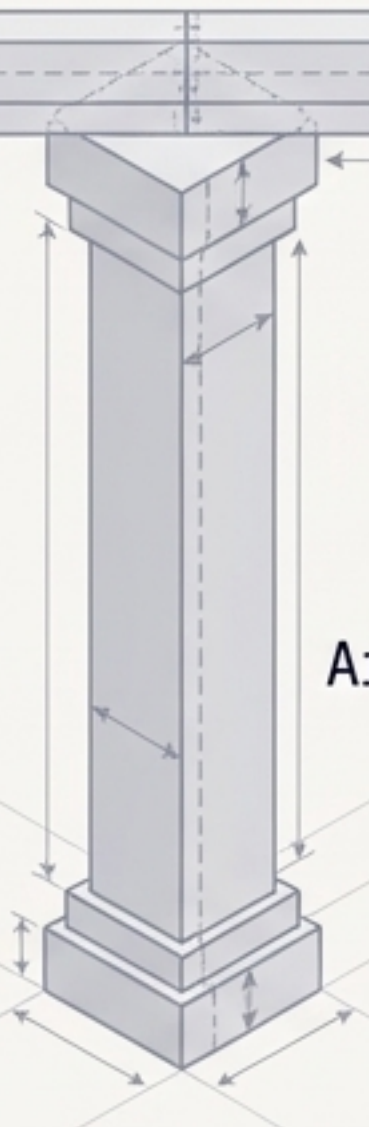
The Horizon: Roadmap & Future Vision

Smart Suggestions

Utilisation Heatmaps



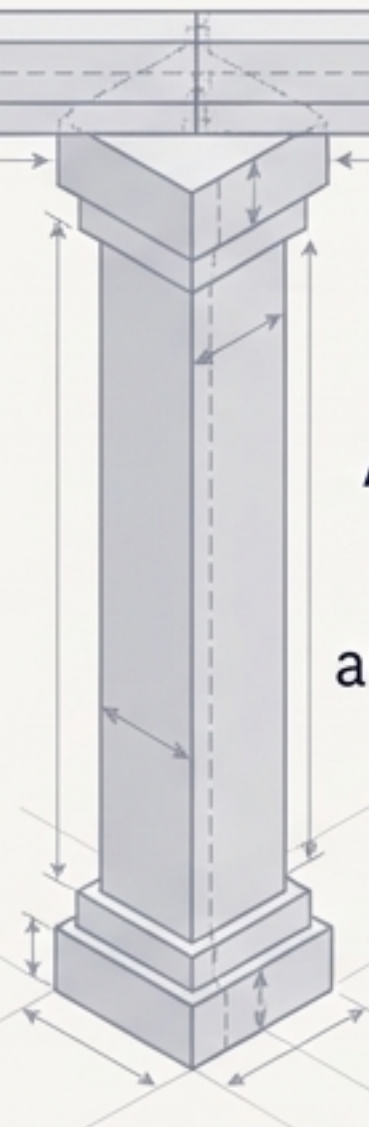
The ParkFlow Impact



A technical diagram of a parking column. It features a square base, a tall central shaft, and a square top. Various dimension lines with arrows indicate measurements for the base width, base height, shaft height, and top width. The column is positioned on a grid of lines.

Maximum Utilisation

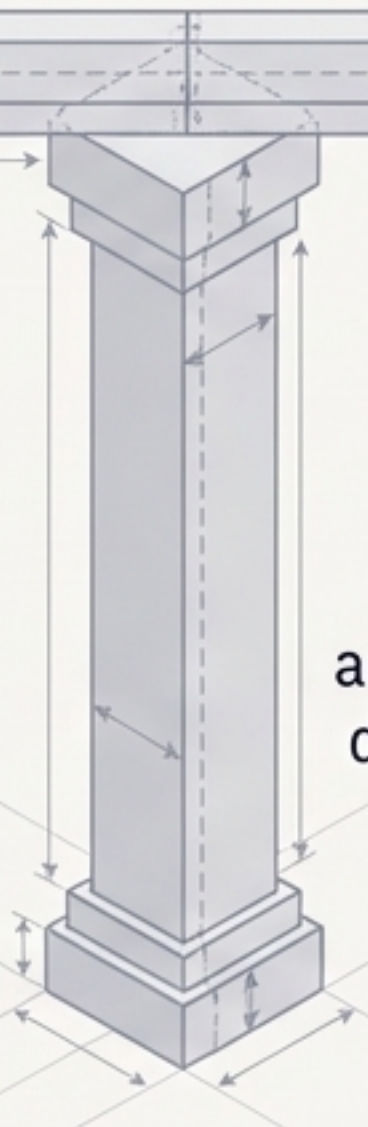
No empty spots.
Artificial scarcity
eliminated.



A technical diagram of a parking column, identical to the first one. It features a square base, a tall central shaft, and a square top. Various dimension lines with arrows indicate measurements for the base width, base height, shaft height, and top width. The column is positioned on a grid of lines.

Zero Administration

Abelium timesheet
automation removes all
manual oversight.



A technical diagram of a parking column, identical to the first two. It features a square base, a tall central shaft, and a square top. Various dimension lines with arrows indicate measurements for the base width, base height, shaft height, and top width. The column is positioned on a grid of lines.

Exceptional Experience

Real-time SSE updates
and interactive CAD maps
delight everyday users.

A flawless operational tool built on an
uncompromising technical foundation.
Seamless for users, zero-touch for admins.